IoT-Enabled Wireless Sensor Network for Dam Safety Management

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***Abstract*— Dams play a vital role in water management, energy production, and flood control. However, their structural integrity and operational safety require continuous monitoring and management. Traditional dam safety systems often involve manual inspections, which can be time-consuming and inefficient. The advent of the Internet of Things (IoT) has revolutionized monitoring and management practices across various industries. This paper explores the development and implementation of a Dam Safety Management System (DSMS) using IoT. The system leverages sensors, cloud computing, and real-time data analytics to enhance dam safety, streamline operations, and mitigate risks. The proposed system aims to provide accurate, timely insights into dam performance, enabling proactive interventions to prevent disasters.**

Keywords*— *IoT, dam safety, wireless sensor networks, real-time monitoring, cloud computing, structural health, early warning systems, data-driven decision-making.

I. Introduction

Dams are critical infrastructures that serve multiple purposes, including irrigation, hydroelectric power generation, and water supply. Despite their importance, dam failures can have catastrophic consequences, including loss of life, environmental damage, and economic disruption. Conventional dam monitoring methods rely heavily on manual inspection and isolated data collection, which often fail to provide a comprehensive and real-time view of dam health. The Internet of Things (IoT) offers a transformative approach to dam safety management. IoT involves the integration of smart sensors, wireless communication, and cloud-based data processing to monitor and analyze various parameters in real time. By deploying an IoT-enabled Dam Safety Management System, authorities can detect potential issues such as structural deformations, seepage, and vibrations at an early stage, thereby enhancing safety and operational efficiency. This paper outlines the architecture, components, and benefits of a DSMS using IoT. Additionally, challenges and future directions are discussed to provide a holistic view of this innovative approach. Rivers and streams play a major role in the livelihoods and cultural practices of Myanmar's predominantly rural populace. In order to construct a dam, the river, stream, and inland areas must be managed for the benefit of human life. In Myanmar, there are almost 200 large dams. Specific types of dams in Myanmar are hydroelectric dams (Yeywa and Myitsone Dam), irrigation dams (Nay Pyi Taw Irrigation Dam) and multipurpose dams (Upper and Lower Paunglaung Dam).

Dams benefit our society greatly, but they can have adverse impacts on the surrounding area. When a dam fails, there will be severe downstream social and economic consequences, including significant losses in terms of people and property. Dam leaks, cracks, and deformation have a major impact on how safety structures operate. The safety monitoring model of the dam based on real time monitoring data is an important means to evaluate the safety of dam operation. The target dam for the proposed system is multipurpose Lower Paunglaung dam. The Lower Paunglaung Dam is situated in Myanmar's Mandalay Division boundary area at latitude 19 º 47' 17" north and longitude 96 º 19' 59". An American company has been studying the possibility of generating hydroelectric power from the Paunglaung River since 1953. The Lower Paunglaung Dam is currently managing dam safety using only manual data. Nowadays, as AI technologies advance, technology and IoT devices are evolving into automated systems. An IoT-based real-time dam water level monitoring and management system is shown in this study. The components of this system include Raspberry Pi 3, seepage sensor, vibration sensor, water level sensor, temperature sensor and WiFi modules. This system collects all real time parameters and weather conditions all around the dam for the prediction of abnormal conditions.

The objectives of this system include to manage water levels in dams and reservoirs to prevent overflows, water scarcity, or mismanagement, to enhance public trust by providing real-time insights and data-driven decisions and to preventing disasters for reducing the financial burden on emergency responses and reconstruction. This paper is structured as follows: related studies are described in Section II. The proposed method is explained in Section III, while Sections IV and V include descriptions of the system's architecture and results. In Section VI, the system is finally concluded.

II. Related works

Some studies presented concerning the dam safety management in real-time monitoring. The authors, Asst. Prof. Shireen Fatima, Parvez Musharaf, Sk Junaid, Shaik Nadeem, Numan Ahmed [1], described an Internet of Things based system for managing and monitoring dam water levels in real time. Sensors and wireless communication technologies send water level data to a central computer for automatic control and remote monitoring. Features like automatic alarms, maintenance prediction, and real-time data analysis stand out. This system uses only one water level sensor and rainfall sensor, so the accuracy is low. In order to protect the public

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from flooding caused by rising water levels in dams and reservoirs, the authors suggested in [3], system that integrates the Internet of Things. Sensor data is gathered and uploaded to a cloud database where automated comparative analytics regarding the rise in water level are recorded. As a result, the public is automatically informed of the earlier phases of the water level rise. Lastly, it was shown that this strategy increases the level of accuracy when compared to the standard monitoring and alerting system method.

In [4], "Iot Based Water Flow Monitoring and Controlling System", the authors suggested a water monitoring and control system for the water utility in order to lessen the existing issue of water waste. The goal of this strategy is to assist utility controllers in enhancing affordable water management systems through the use of emerging technologies, including the Internet of Things. The information gathered from every sensor is utilized for analysis in the distribution of water supplies. IOT services that examined distant water lines, improve their monitoring process, and enable real-time controlling and sufficient data processing have lately increased interest in water flow monitoring and controlling systems.

In [5], the authors investigated how to use IoT technology to transform a manual monitoring system into a smart environment. They presented a thorough methodology for managing and monitoring dam safety that was based on widely used technology. The results demonstrated that incorporating new technology into the dam monitoring procedure can enhance overall performance and lower human error. This study showed how human error can be reduced in manual dam monitoring by utilizing RFID and Bluetooth. These results are intended to assist stakeholders in understanding the advantages of utilizing BT and RFID. The study shows that with low-cost expense, it is feasible to improve the current monitoring system and acquire higher-quality data.

Hao Tang, "A Review of Safety Inspection Methods for Reservoir Dams", [6] discussed about several techniques for monitoring dam displacement and detecting leaks, all of which are part of the overall safety inspection of reservoir dams. He presented four methods for detecting leakage in reservoir dams. They are drilling method, seepage field method, computed tomography method (CT method) and applicability analysis of seepage detection methods. Locations of geophysical anomalies or suspected leaks are the sites of drilling. To determine the seepage flow field in the area, inverse fitting analysis is performed by monitoring the groundwater flow velocity and flow field in boreholes at various sites. The CT approach reconstructs images of the elastic and electrical distribution patterns within the measured rock mass by using ray scanning and inverse calculation of the information gathered. The methods for detecting leaks in reservoir dams are based on the theories of widely used leak detection techniques and take into account the structural features of concrete, rockfill, and earth fill dams. They concentrate on locating leak entrance points and leakage routes. He reviewed various methods for dam safety and management systems.

In [7], the authors identified the main issues with dam safety and stated the necessity of a more sophisticated and integrated monitoring system. The suggested framework included real-time data processing, sensor networks, and predictive analytics to facilitate early warning systems and proactive risk assessment. The float sensor provides alert information, and the ultrasonic sensor determines the appropriate water level and shows it on a 16x2 LCD display. Send a text message when the dam's water level reaches its full and over the threshold value. The results of this study support the continued efforts in the field of dam safety management by urging the use of cutting-edge monitoring technologies to guarantee safe and robust infrastructure.

Nikhil M. Dhandre, P. D. kamalasekaran and Pooja Pandey, "Dam Parameters Monitoring System", [8] established a web portal that will track and display real-time weather and dam parameters, such as water level, precipitation, gate position, temperature, humidity, and more. In addition, this system created GUI software that offers two different kinds of program operational facilities, namely, autopilot and manual data mode. The software gathers parameter information from the relevant sensors and stores it in a database at the back end. The data that has been spilled can be used to host web portals and make additional decisions. This suggested strategy mainly uses the Internet of Things to enable data exchange through web databases. the data management and use will also be enhanced, and data sharing to distant systems will be facilitated.

In [9], several sensors are used in the preventive maintenance system to check out various dam components. At that, measure the data from the relevant sensors and submit it to the server. Look for signs of internal erosion within the dam, corrosion on the gate dams, wall cracking, pipe deterioration, and high turbidity in the water downstream of the dam. The information gathered can be utilized for additional decision-making and Android applications. The process basically uses the Internet of Things to facilitate data sharing. The system gave SMS and alarm/alert capabilities to dam authority. IoT can reduce the time lag caused by human data observation and transmission, which reduces the chance of losing real-time data.

In [11], the authors suggested that the use of artificial intelligence and the Internet of things for maintenance has been studied, as have technologies for the monitoring and safety management of the dam structures. However, measurement mistakes might occur throughout the communication and measurement procedures of sensors utilized in IoT technology. To solve these problems, technologies for accurate measurement result interpretation and effective communication are needed. This study used AIoT (AI+IoT) to design a system for dam safety management and communication environment optimization, and to improve the convenience and safety management of dam monitoring for fill dams, where heavy rainfall regularly causes damage and traditional safety management technologies are used for monitoring, which have a number of drawbacks.

In this proposed system, the WSN model will be used to incorporate the data and information gathered from sensors regarding the environmental condition of the dam region. Furthermore, cloud storage will be used to store the sensor data. After that, the dam authority will be informed if the threshold values are exceeded. The working approach to monitor the numerous factors related to dam maintenance and to maintain dam safety is presented in this study.

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III. PROPOSED METHODOLOGY

Wireless Sensor Networks (WSN) significantly enhance dam safety management by providing real-time, continuous monitoring of critical parameters. This system can regularly and closely monitor the dam's parameters. Using WSN technology, the sensor data is wirelessly sent to a central system. Collected data is uploaded to a cloud platform for storage and analysis in Figure 1. Real-time data detect by the sensors and compare to the predetermined threshold value. If the sensor value is greater than threshold value, alert message is sent to the dam authority person or organization.

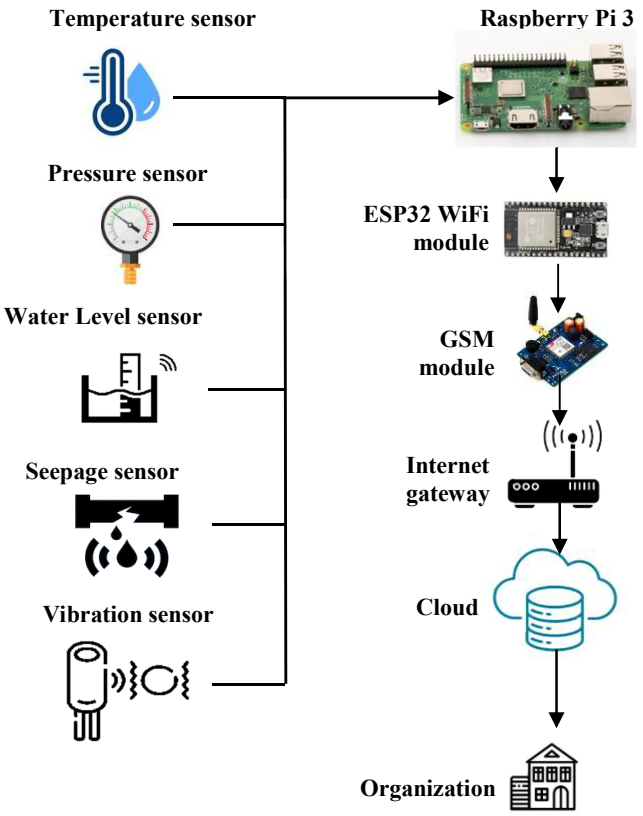

Figure 1. The architecture of the system

A. *Sensor Network*

- **Temperature sensor**: For a dam to monitor environmental conditions and maintain the structural integrity of the dam and its surroundings, a temperature sensor is essential. Sensors are positioned closer to the surface to monitor the effect of external temperature variations due to weather conditions, seasonal changes, or temperature gradients from the water in the reservoir.

- **Pressure sensor:** A dam's water pressure sensor is a vital component of equipment that measures the force that the water exerts at different locations along the dam, including the base, inside the body, and on the surface. These sensors play a key role in ensuring the safety and structural integrity of dams by monitoring water levels, detecting potential leaks or cracks, and helping in overall dam management. Pressure sensors are installed along the upstream and downstream faces of the embankment to measure the external pressure exerted by water in the reservoir or the groundwater.

- **Water level sensor:** A dam's water level sensor is an essential tool for measuring and tracking the water level in a reservoir or dam. These sensors help to the dam's safety by giving real-time water level data, which is necessary for preventing overflow, and efficiently managing water resources. These sensors are placed along the upstream face of the dam, which faces the reservoir.

- **Seepage sensor:** Seepage sensors are designed to detect the presence of water that passes through or under the dam, helping engineers identify issues before they become serious threats to the dam's integrity. Monitoring seepage is essential to dam safety because water seeping through a dam's foundation or construction may indicate possible erosion, leaks, or structural issues. Seepage sensors are installed near the spillway, outlet works, or other water release structures.

- **Vibration sensor:** For dam safety and structural health monitoring, vibration sensors are necessary. These sensors can detect vibrations caused by factors such as water movement, seismic activity, or internal structural changes. Vibration monitoring can help detect problems like internal erosion, settling, or cracks early on, preventing serious damage and guaranteeing the integrity of the dam. These sensors are placed along the dam's crest, which is the topmost portion of the structure.

B. *Wireless Communication Network*

- The sensor data is transmitted wirelessly to a central system using technologies such as LoRaWAN, Zigbee, or 5G. The network ensures low-latency, energy-efficient data transmission over long distances.

C. *Data Collection and Transmission*

- **Data Acquisition:** Implement a system that collects sensor data continuously. Acquire sensor data and transmit it wirelessly to a central system.

D. *Cloud-Based Data Storage and Processing*

Collected data is uploaded to a cloud platform for storage and analysis. The cloud infrastructure facilitates:

- **Data Aggregation:** Organizing data from various sensors.

- **Real-Time Analysis:** The system may alert users to abnormal levels or potential issues so that timely action can be taken. It might improve dam safety and management by using data analytics to predict and prevent dam problems.

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- **Remote Monitoring:** Allow remote system monitoring for dam operators. This allows for rapid emergency and anomaly responses.

- **Alarm System:** Set threshold-based alarms to warn dam management immediately for dangerous conditions.

- **Threshold Values:** Values are predetermined thresholds for sensor data, over which the limit rate is deemed hazardous or abnormal. These values are determined based on: historical data, engineering analysis of the dam's design and industry safety standards.

E. User Interface and Alerts

A web-based dashboard and mobile application provide an intuitive interface for monitoring dam health. The system issues automated alerts via SMS, email, or app notifications in case of abnormal readings.

IV. System Architecture

Maintaining the dam is necessary to keep it safe, in good working stability, and to increase its lifespan. The proposed strategy enables early warning and monitoring for the dam safety management system.

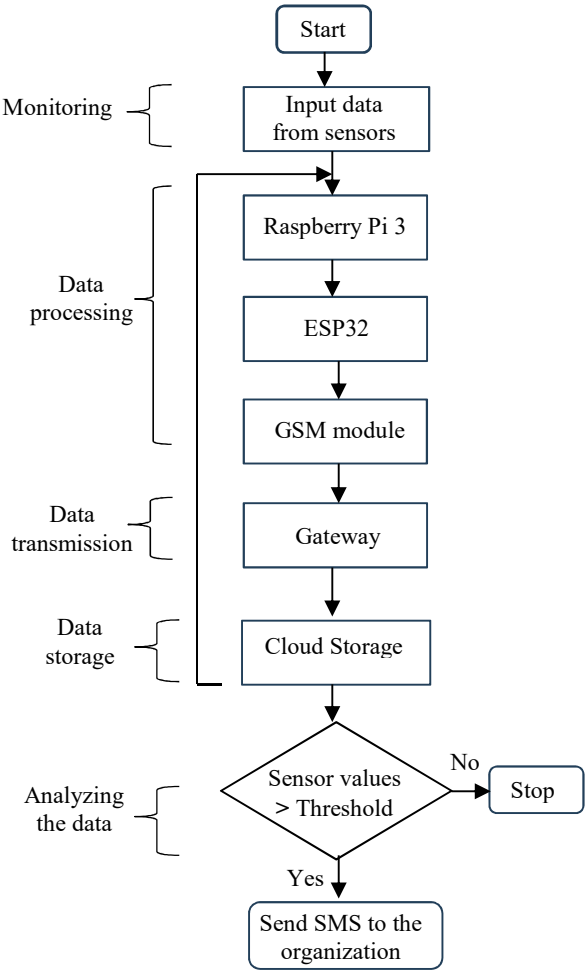

Figure 9. The system's flow structure

In Figure 9 describes how the safety management system for real-time monitoring and warning at the Lower Paunglaung dam is being implemented:

- First, sensors will be installed around the dam site.
- Second, the Raspberry Pi is connected to all of the sensors.
- Third, the Raspberry Pi receives the sensor data and sends it to the WiFi module (ESP32).
- Fourth, data flows through the internet gateway that connects to the cloud's data storage.
- Finally, the output result will be sent to the organization via the cloud.

V. Results and Discussion

The dam's safety monitoring data acts as an indicator of its current operational state. It is an essential for evaluating the safety status of dams and has powerful real-time capability to identify anomalies in the dam as soon as they occur. Traditional dam monitoring systems have mostly depended on routine evaluations and manual inspections. A real-time monitoring strategy for dams is required due to the growing complexity of environmental conditions and the possible hazards caused by weather. The water level of a dam is influenced by a variety of factors, ranging from the weather condition of the dam site. Rainfall has the greatest effect on the water level in a reservoir behind a dam. Heavy rain increases the flow of water into the dam, raising the water level, while periods of drought or low rainfall can cause the water level to drop. Additionally, the water level in the reservoir depends on how much water enters the dam from rivers or streams. A dam's water level is affected by a complicated interaction between natural elements (such rainfall, snowmelt, and evaporation) and man-made ones (like flood control, hydropower operations, and inflow/outflow management). In order to maintain water levels within safe and ideal limits for dam safety, environmental sustainability, and energy generation, real-time monitoring and modifications are necessary for effective water level control. This proposed system will be selected and tested at Lower Paunglaung Dam in Nay Pyi Taw, Myanmar.

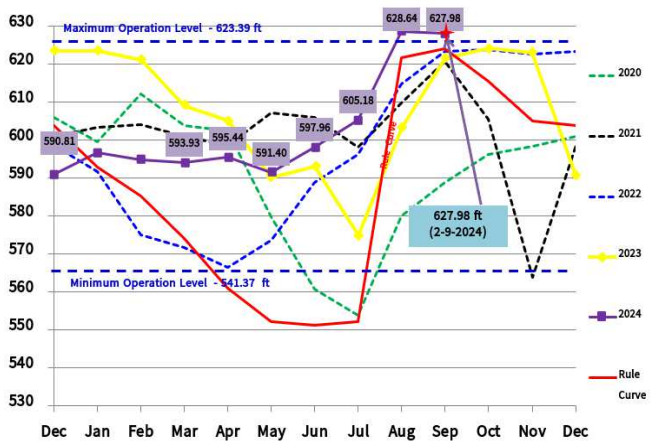

Figure 10. Reservoir water level graph of Lower Paunglaung Dam

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The Lower Paunglaung Dam's historical data of actual monthly water level for the year of 2020 to 2024 is shown in Fig. 10. In Myanmar, August and September are the months with the most rainfall in every year. In September 2024, heavy rains in Myanmar caused widespread flooding, causing loss of life and property. Additionally, the dam's reservoir's water level reached its maximum in rainy season. The dam may break if the reservoir has an excessive amount of water. Therefore, the water level of the dam should be monitored in real time. When the seepage rate exceeds the threshold, it typically suggests a risk of erosion, cracks, or internal damage that could compromise the dam's structural integrity. The results of this system include real-time monitoring of the dam, giving early warnings when dangerous conditions occur, reducing human labor due to the use of real-time automatic systems, and the system's accuracy is high due to the use of sensors to collect data.

VI. CONCLUSION

Dams are very important for human daily life, providing water supply for cities, producing electricity, and protecting against natural disasters such as floods and droughts. As IoT technology advances, using technology to monitor the safety of dams has become increasingly important. A real-time dam safety system offers numerous benefits, primarily focused on improving the safety, efficiency, and management of dam infrastructure. Long-term monitoring of dam functioning is made possible by sensors that continuously collect data. By detecting issues early, a real-time dam safety system helps prevent catastrophic failures, which could be extremely costly in terms of repairs, environmental damage, and lost lives. Predictive maintenance is made possible by the system, which recognizes areas that need repair before they become serious issues. Automation reduces the need for manual inspections, which can be labor-intensive and prone to human error. While real-time monitoring systems for dam safety management may require a higher upfront investment, but they offer substantial cost-saving benefits in the long term. Sensors in real-time monitoring systems measure vital characteristics including temperature, vibrations, seepage, pressure, and water levels continually. For dam safety management, real-time monitoring systems offer more objective, reliable, and accurate data than manual inspections. This system will provide SMS alert facilities for dam authorities when abnormal conditions occur. Additionally, it will enhance data administration, use and facilitate data sharing to remote systems.

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